

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – IX
PAPER –2
TEST DATE: 17-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

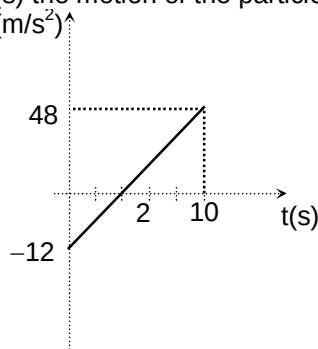
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Two stars S_1 and S_2 of mass M each are moving in circular orbits about their common centre of mass 'O' due to mutual gravitational force of attraction. The separation between them is $4R$. An asteroid of mass m ($m \ll M$) is in a coplaner circular orbit about same centre 'O'. It is observed that the asteroid is also moving in a circular orbit with same angular speed. Neglect size of stars and asteroid compared with R . Then choose the correct option(s).

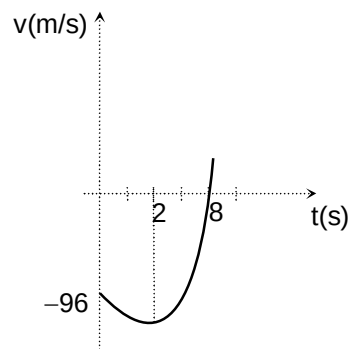
- (A) Time period of revolution of stars and asteroid is $8\pi\sqrt{\frac{2R^3}{GM}}$
- (B) Radius of the circle traced by asteroid is $2\sqrt{3}R$
- (C) The total energy of the asteroid is $\frac{-5GMm}{16R}$
- (D) Radius of the circle traced by the star S_1 is $2R$.

2. The position of a particle which moves along a straight line is defined by the relation $x = t^3 - 6t^2 - 96t + 37$, where x is expressed in meters and t is in seconds. Which of the graph(s) represent(s) the motion of the particle?

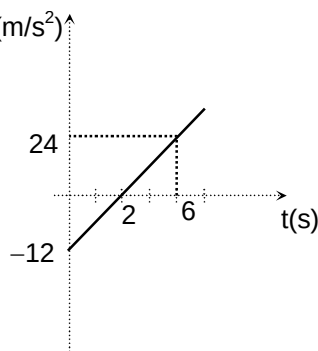
(A) $a(m/s^2)$



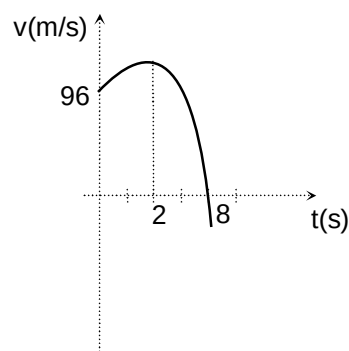
(B) $v(m/s)$



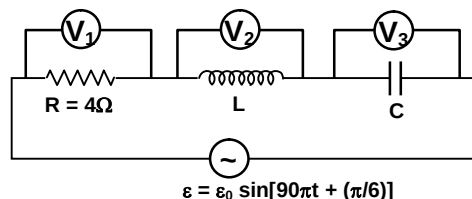
(C) $a(m/s^2)$



(D) $v(m/s)$



3. A flat circular annular disc of mass M , outer radius a and inner radius b . The measured value of M , a and b are, $M = (0.30 \text{ kg} \pm 0.0003 \text{ kg})$, $a = (4 \text{ cm} \pm 0.001 \text{ cm})$ and $b = (2 \text{ cm} \pm 0.001 \text{ cm})$. Then,
- Percentage error in the calculation of a and b are in ratio 1 : 2.
 - Percentage error in the calculation M is 0.1%
 - Percentage error in the calculation of moment of inertia of the disc about an axis passing through the centre and perpendicular to the plane of disc is 0.16%
 - Percentage error in the calculation of moment of inertia of the disc about an axis passing through the centre and perpendicular to the plane of disc is 1.6%
4. When a yellow light of wavelength $6000 \text{ \AA}$ from a sodium lamp falls on a certain photocell, a negative potential of 0.30 volt is needed to stop all the electrons from reaching the collector. Then choose correct option(s).
- Work function of the given photocell is 2.5 eV (approximately).
 - Work function of the given photocell is 1.78 eV (approximately).
 - A negative potential of 0.6 Volt will be needed to stop all the electrons if light of wavelength $\lambda = 4000 \text{ \AA}$ is used
 - A negative potential of 1.32 Volt (approximately) will be needed to stop all the electrons if light of wavelength $\lambda = 4000 \text{ \AA}$ is used
5. 10g of ice at 0°C is mixed with 5g of steam at 100°C . If latent heat of fusion of ice is 80 cal-g^{-1} and latent heat of vaporization of 540 cal-g^{-1} . When thermal equilibrium is reached then which of the following is/are correct?
- Temperature of the mixture is 0°C
 - Temperature of mixture is 100°C
 - Mixture contains 13.3 g of water and 1.67 g of steam
 - Mixture contains 5.3 g of ice and 9.7 g of water
6. In the figure shown the reading of voltmeter are $V_1 = 80 \text{ V}$, $V_2 = 90 \text{ V}$ and $V_3 = 30 \text{ V}$. Then choose the correct option(s).



- The peak value of current is $20\sqrt{2}$ ampere.
- The peak value of emf is $100\sqrt{2}$ volts.
- The value of L , is $\frac{1}{20\pi} \text{ H}$
- The value of C for which the circuit will produce maximum heat is $\frac{1}{405\pi} \text{ F}$

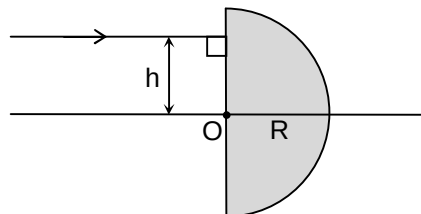
Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

A semi-cylinder made of a transparent plastic has a refractive index of $\mu = 5/3$ and radius of $R = 30$ cm lies in vacuum. There is a narrow incident light ray perpendicular to the flat side of the semi-cylinder at height 'h' from the axis of symmetry. (take $\tan 37^\circ = 3/4$)

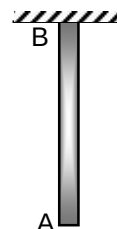
Now answers the following questions.



7. What is the maximum value of 'h' so that the light ray can still leave the other side of the semi-cylinder?
 - (A) 12 cm
 - (B) 8 cm
 - (C) 18 cm
 - (D) 24 cm
8. When the refractive index of transparent plastic is $\mu = \sqrt{2}$, the value of 'h' is chosen such that total internal reflection just takes place, then time for which light remains inside the cylinder, is
 - (A) 10^{-9} s
 - (B) 2×10^{-9} s
 - (C) 4×10^{-9} s
 - (D) 8×10^{-9} s

Paragraph for Question Nos. 09 and 10

Consider a rope AB of non-uniform density as shown in the figure. If we consider direction A to B as the direction of increasing y, with $y = 0$, at A. The linear mass density of rope is given by $\mu = \left(\frac{y}{\ell}\right)^3 \mu_0$, where μ_0 is a positive constant and ℓ is length of the rope. Now one pulse is produced at the lower end A. Then answer the following questions



9. The time to move the pulse from A to B.
 - (A) $\sqrt{\frac{\ell}{g}}$
 - (B) $2\sqrt{\frac{\ell}{g}}$
 - (C) $3\sqrt{\frac{\ell}{g}}$
 - (D) $4\sqrt{\frac{\ell}{g}}$

10. Acceleration of the pulse when $y = \ell/4$.

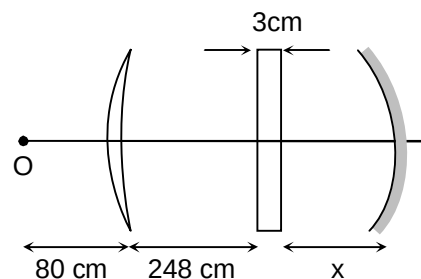
(A) $g/4$
 (B) $g/2$
 (C) g
 (D) $g/8$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. Two rods each of length L_1 and one rod of length L_2 form isosceles triangle having base of length L_2 . Co-efficient of linear expansion of base is $4 \times 10^{-6} / ^\circ\text{C}$ while for rod of length L_1 is $25 \times 10^{-6} / ^\circ\text{C}$. Find the ratio $\left(\frac{L_2}{L_1}\right)$, if the length of latitude remains same when the temperature of the system is increased by 10°C .

12. A concavo-convex lens made of glass of refractive index 1.5 has surfaces of radii 20 cm and 60 cm. A concave mirror of radius of curvature 20 cm is placed co-axially to the lens. A glass slab of thickness 3 cm and refractive index 1.5 is placed close to the mirror in the space between the mirror and lens as shown in the figure. The distance of the nearest surface of the slab from lens is 248 cm. An object is placed 80 cm to the left of the lens. If the final position of the image formed after the refraction from lens, refraction from slab, reflection from mirror, refraction from the slab and lens is at the object O, find the distance x (in cm) of the mirror from nearer surface of slab.



13. A train moving towards a hill at a speed 144 km/hr, sounds whistle of frequency 300 Hz. Wind is blowing from the hill with a velocity 360 km/hr at an angle of 53° with the railway track. If the speed of sound in air is $\frac{400}{3}$ m/s, then find the frequency (in Hz) heard by a man on the hill.

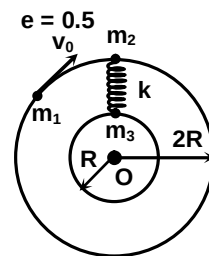
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

Three particles each of mass ' m ' can slide on fixed frictionless circular track in the same horizontal plane as shown in the figure. Particle m_1 moves with velocity v_0 and hits particle m_2 , the coefficient of restitution being $e = 1/2$. Assume m_2 and m_3 are initially in rest and lie along a radial line before impact and the spring is initially unstretched. The velocity of m_3 when extension in the spring is maximum, is X m/s and the maximum extension in the spring is Y meter. (Take $v_0 = 20$ m/s and $\frac{m}{k} = 5 \text{ kg/(N/m)}$)



14. The value of X is

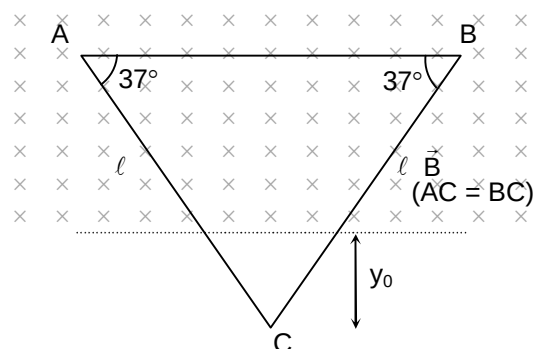
15. The value of Y is

Question Stem for Question Nos. 16 and 17

Question Stem

A current carrying loop is in the shape of isosceles triangle of mass m lies in the vertical plane. There exists uniform horizontal magnetic field B in the region as shown in the figure. The current in the loop is i . The value of y_0 is X meter when loop is in equilibrium. If the loop is displaced slightly in its plane perpendicular to its side AB from its equilibrium position and released, then time period of its oscillation is Y sec. (neglect the

induced emf in the loop, take $\frac{mg}{iB} = \frac{80}{3}$, $\pi = 3.14$ and $g = 10 \text{ m/s}^2$)



16. The value of X is.....

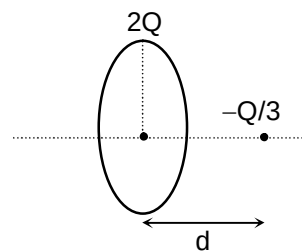
17. The value of Y is.....

Question Stem for Question Nos. 18 and 19

Question Stem

A ring of mass $2M$, radius R and charge $2Q$ is placed as shown in the figure. A point charge $-\frac{Q}{3}$ and mass M is placed on the axis of ring at a distance d from the centre of the ring as shown in the figure. ($d \ll R$). When system is released it performs oscillations. The energy of oscillation of the system is

$\frac{Q^2}{m\pi\epsilon_0 R^n}$ and the time period of oscillation is $T = \sqrt{\frac{X\pi^3\epsilon_0 MR^3}{YQ^2}}$.



18. The value of $|m - n|$ is

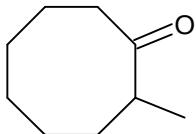
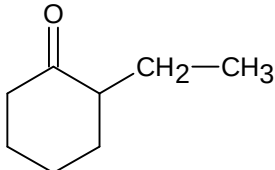
19. The value of $|X - Y|$ is

Chemistry

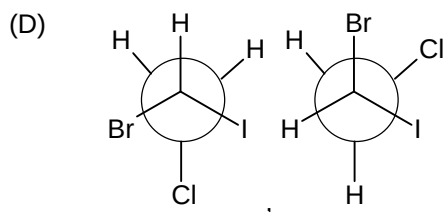
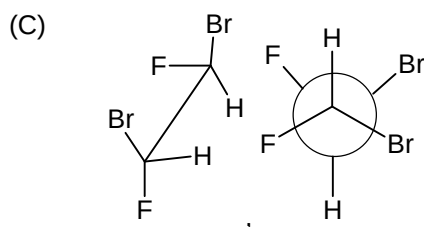
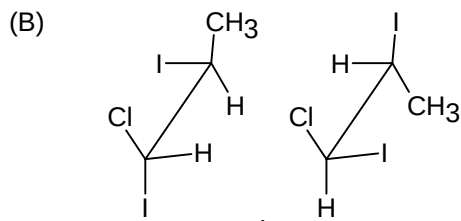
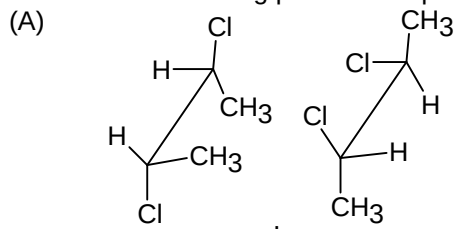
PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. Among the following intensive properties are
 (A) E_{cell}°
 (B) E_{cell}
 (C) Normality
 (D) Entropy
21. Which of the following solution is/are hypertonic at constant T with 0.4 M NaCl showing 80% dissociation?
 (A) 0.7 M glucose
 (B) 0.3 M Na_3PO_4 showing 90% dissociation
 (C) 1 M $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ showing 30% association
 (D) 0.5 M MgCl_2 showing 20% dissociation
22. In isothermal compression of ideal gas
 (A) W is +ve
 (B) ΔH is zero
 (C) ΔS_{gas} is – ve
 (D) ΔE is zero
23. Tautomer of which of the following can show geometrical isomerism
 (A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
 (B) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
 (C) 
 (D) 

24. Which of the following pairs of compounds is/are identical?



25. Which set of quantum numbers is/are not consistent with the theory?

- (A) $n = 2, \ell = 0, m = -1, s = +\frac{1}{2}$
- (B) $n = 2, \ell = 0, m = 0, s = -\frac{1}{2}$
- (C) $n = 1, \ell = -1, m = 0, s = -\frac{1}{2}$
- (D) $n = 3, \ell = 3, m = -3, s = +\frac{1}{2}$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

An orange solid (A) on heating gives a green residue (B), a colourless gas (C) and water vapours. The dry gas (C) on passing over heated Mg gave a white solid (D). (D) on reaction with water gave a gas (E) which formed black precipitate with mercurous nitrate solution.

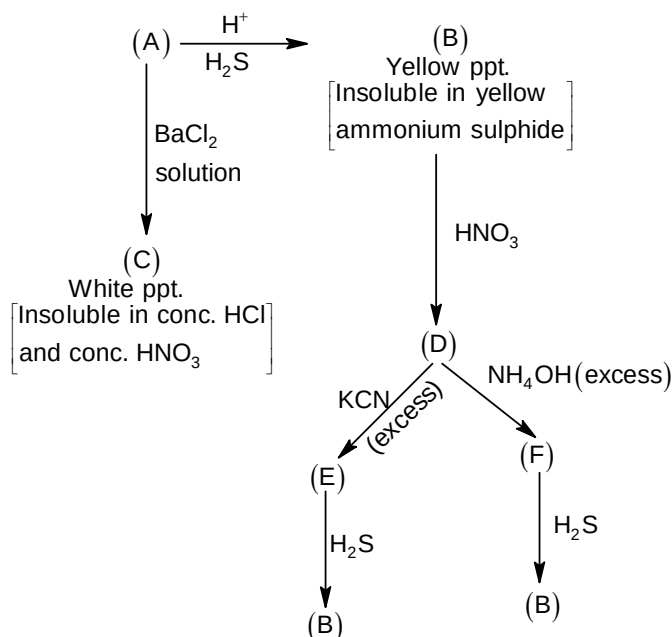
26. Select the incorrect statement

- (A) The central atom (s) of the anion of solid (A) has sp^3 hybridization.
- (B) The orange solid (A) is diamagnetic in nature.
- (C) The anion of orange solid (A) is oxidizing in nature.
- (D) All metal oxygen bond lengths are equal in anion of solid (A).

27. Which of the following is false for the gas (E)?

- (A) It gives a deep blue colouration with CuSO_4 solution.
- (B) It is oxidized to a colourless gas (neutral oxide) at 1200 K in presence of a catalyst Pt/Rh in air.
- (C) It gives the same gas (C) with potassium permanganate solution.
- (D) It gives black precipitate with HgCl_2 .

Paragraph for Question Nos. 28 and 29



28. What is "C"?

- (A) PbCl_2
- (B) BaSO_4
- (C) BaSO_3
- (D) PbSO_4

29. What is "B"?
- (A) CdS
- (B) PbCrO_4
- (C) HgO
- (D) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. Number of co-ordination isomerism for the complex $\left[\text{Pt}(\text{NH}_3)_4\right]\left[\text{PtCl}_4\right]$ is (including itself)
31. Number of stereo isomers in the complex $[\text{Ma}_2\text{b}_2\text{c}_2]$ [where a, b, c are monodentate ligand]
32. The following equilibrium are established in mixing of two gases x_2 and y in sealed container at constant temperature T.
- $$3\text{x}_2(\text{g}) \rightleftharpoons \text{x}_6(\text{g}), \quad K_p = 1.6 \text{ atm}^{-2}$$
- $$\text{x}_2(\text{g}) + \text{y}(\text{g}) \rightleftharpoons \text{x}_2\text{y}(\text{g}), \quad K_p = z \text{ atm}^{-1}$$
- When $\text{x}_2(\text{g})$ and $\text{y}(\text{g})$ are mixed in 2 : 1 molar ratio, the total pressure of gases at equilibrium is found to be 1.4 atm and partial pressure of $\text{x}_6(\text{g})$ is 0.2 atm. The value of 6W is.

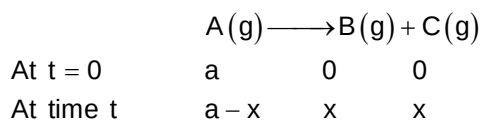
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

A reaction is said to be first order if it's rate is proportional to the concentration of reactant. Let us consider a reaction



The rate of reaction is given by the expression $\frac{dx}{dt} = k(a - x)$ and integrated rate equation for given reaction is represented as $k = \frac{1}{t} \ln\left(\frac{a}{a - x}\right)$ where a = initial concentration and (a – x) = concentration of A after time t.

33. Thermal decomposition of compound A is a first order reaction. If 75% of A is decomposed in 100 min. How long will it take for 90% of the compound to decompose? [Given: $\log 2 = 0.30$]

34. Consider a reaction $A(g) \longrightarrow 3B(g) + 2C(g)$ with rate constant $1.386 \times 10^{-2} \text{ min}^{-1}$. Starting with 2 moles of A in 12.5 litre vessel initially, if reaction is allowed to take place at constant pressure and at 298 K then find the concentration of B after 100 min? $\left[e^{-1.386} = 0.25 \right]$

Question Stem for Question Nos. 35 and 36

Question Stem

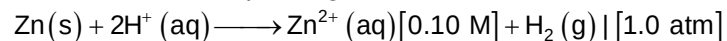
0.257 g of an organic compound is analysed for nitrogen by Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.1 M HCl which required 23.2 mL of $\frac{N}{10}$ NaOH for the complete neutralization.

35. What is percentage of 'N' in the compound?
36. In Carius method of estimation of halogen, 0.15 g of an organic compound gave 0.12 g of AgBr. What is % of Bromine in the compound?
[Take : $M_{\text{AgBr}} = 188 \text{ g/mol}$; $M_{\text{Br}} = 80 \text{ g/mol}$]

Question Stem for Question Nos. 37 and 38

Question Stem

EMF of a cell corresponding to the reaction:



is 0.61 V at 25°C. Also, at 25°C, $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$. $\left[\frac{2.303RT}{F} = 0.06 \right]$

37. The pH of solution in cathode chamber is.....
38. ΔG of the above cell reaction is y, then the value of |y| in kJ? (Given $1F = 96500$)

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. If the value of limit ; $\lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \sum_{k=2}^i \frac{k-1}{k!} \right)$ is L, then (where $[.]$ denotes greatest integer function)
- (A) $\left[\frac{[L] + 2L}{[L]} \right] = 3$
- (B) $\left[\frac{[L] + 2L}{[L]} \right] = 4$
- (C) $\left[\frac{[L] + [L-1]}{2[L]} \right] = 0$
- (D) $\left[\frac{[L] + [L-1]}{2[L]} \right] = 1$
40. Suppose that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the relation $f(x + g(y)) = 2x + y + 5$, then if function $g(x + f(y)) = k_1(x) + k_2(y) + k_3$, then
- (A) $2k_1 + k_2 = 1$
- (B) $2k_1 + k_2 = 2$
- (C) $k_3 - k_1 = 0$
- (D) $k_3 - k_1 = 2$
41. ${}^mC_1 \cdot {}^nC_m - {}^mC_2 \cdot {}^{2n}C_m + {}^mC_3 \cdot {}^{3n}C_m \dots$ is equal to
- (A) coefficient of x^n in $(1 - (1 + (1 - x)^n)^m)$
- (B) coefficient of x^n in $(1 - (1 - (1 + x)^n)^m)$
- (C) coefficient of x^m in $(1 - (1 - (1 + x)^n)^m)$
- (D) coefficient of x^m in $(1 - (1 + (1 - x)^n)^m)$
42. Given an alphabet with three letters a, b, c, then the number of words of n letter which contain even number of a's?
- (A) $2^n \left(\left(1 + \frac{1}{2} \right)^n + \left(1 - \frac{1}{2} \right)^n \right)$
- (B) $2^{n-1} \left(\left(1 + \frac{1}{2} \right)^n + \left(1 - \frac{1}{2} \right)^n \right)$
- (C) $\frac{1}{4}(3^n + 1)$
- (D) $\frac{1}{2}(3^n + 1)$

43. For each integer $n \geq 1$. Let $a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \log n$ and $b_n = a_n - \frac{1}{n}$, then
- (A) $a_{n+1} > a_n$
 (B) $a_{n+1} < a_n$
 (C) $b_{n+1} > b_n$
 (D) $b_{n+1} < b_n$
44. Let point A is taken on line $x + y - 9 = 0$ and B and C are taken on circle $2x^2 + 2y^2 - 8x - 8y - 1 = 0$ such that line AB passes through centre of circle and $\angle BAC = 45^\circ$, then range of x coordinate of point A is superset of which of the following?
- (A) $[3, 4]$
 (B) $[2, 4]$
 (C) $\left[\pi, \frac{3\pi}{2}\right]$
 (D) $[\pi, 5]$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

N dice are rolled once, then the probability of getting sum t is expressed as $P_d(t, N)$. Where dice is a d sided dice with face marked 1, 2, 3, 4, d.

45. Which of the following is an **incorrect** combination?
- (A) $P_6(14, 3) = P_6(14, 5)$
 (B) $P_3(5, 2) = P_3(5, 3)$
 (C) $P_3(10, 4) = P_3(10, 5)$
 (D) $P_4(9, 3) = P_4(9, 4)$
46. The value of $P_9(15, 2)$ is equal to
- (A) $\frac{4}{71}$
 (B) $\frac{4}{81}$
 (C) $\frac{8}{81}$
 (D) $\frac{8}{71}$

Paragraph for Question Nos. 47 and 48

Let consider lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $2x + 5y - z + 47 = 0 = 2x + y + z + 7$

47. The shortest distance between both given line is
 (A) $2\sqrt{30}$
 (B) $3\sqrt{30}$
 (C) $\sqrt{30}$
 (D) $\sqrt{75}$
48. Equation of plane parallel to both of lines and which lies midway between the lines
 (A) $2x + 5y - z + 2 = 0$
 (B) $2x - 5y + z - 2 = 0$
 (C) $2x + 5y - z - 2 = 0$
 (D) $2x - 5y - z - 2 = 0$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. Let R be the circum-radius of triangle ABC and let G and H be its centroid and orthocentre respectively. Let F be the midpoint of GH, then if $AF^2 + BF^2 + CF^2 = kR^2$, then k is equal to
50. In the coordinate plane, the area of the region bounded by the line passing through (1, 2) with the slope α and the parabola $y = x^2$ is denoted by $S(\alpha)$ where α varies in the range of $0 \leq \alpha \leq 6$, then find the value of α such that $S(\alpha)$ is minimized
51. Given a triangle ABC, let a, b, c denotes the length of its sides and m, n, p the length of its medians. For every positive real α , let $\lambda(\alpha)$ be the real number satisfying $a^\alpha + b^\alpha + c^\alpha = (\lambda(\alpha))^\alpha \{m^\alpha + n^\alpha + p^\alpha\}$, then of value of $\lambda(2)$ is k, then $[k]$ is (where $[.]$ denotes the greatest integer function)

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

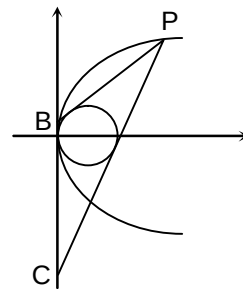
If the abscissa and the ordinate of on equilateral triangle are the roots of $a_1x^3 + b_1x + c_1 = 0$ and $a_2x^3 + b_2x + c_2 = 0$, then if $k_1 \left[\frac{b_1}{a_1} + \frac{b_2}{a_2} \right] + k_2R^3 = 0$ (where R is the circumradius of triangle ABC); k_1 and k_2 are coprime to each other), then

52. Value of k_1 is
53. Value of k_2 is

Question Stem for Question Nos. 54 and 55

Question Stem

As is shown in figure, $P(x_0, y_0)$ is a moving point on the parabola $y^2 = 2x$, point B and C are on y-axis, and the sides of triangle PBC touches the circle $(x - 1)^2 + y^2 = 1$.



54. The minimum value of area of triangle PBC is _____ (in sq. unit)
55. The value of x_0 when the area of $\triangle PBC$ is minimum

Question Stem for Question Nos. 56 and 57

Question Stem

The largest area of rectangle whose

56. Base is on x-axis and upper two vertices lies on parabola $y = 12 - x^2$
57. Base is along with x-axis and inscribed in equilateral triangle having base as x-axis and side length 20 unit